

Admission

The agent sees the history at presentation: present illness; past, social and family history.

Hidden diagnosis
 $y \in \mathcal{D} \cup \{\text{OTHER}\}$

$\mathcal{D}$: appendicitis, cholecystitis, diverticulitis, pancreatitis

Agent π_θ

Qwen3-8B + LoRA;
reasons, then emits one
action per turn.

finding o_t

ASK, TEST(x)

Patient record

Returns what was recorded
for this admission; nothing
is generated.

ASK(exam) is free; TEST(x)
orders one of 22 tests at its
price; at most 8 investigations.

COMMIT(d, p)

names a diagnosis d with
a stated probability p .

DEFER(q , reason)

hands the case to a
clinician with a ranked,
probabilistic differential q .

Scored in clinical units

$$R = \alpha 1[d=y] - \lambda (p - 1[d=y])^2 \\ - \kappa C[y, d] - c \sum_t \text{cost}(a_t)$$

accuracy, Brier score, severity of the error,
test cost

Credited against the counterfactual

$A = E - \hat{V}(s)$: the value of escalating
minus the value of not escalating, from
the same state (Fig. 2).

The handoff q is scored with a proper
scoring rule.

Evaluated on closed world: MIMIC-IV-Ext-CDM val + test (480) · open world: OTHER from groups seen in training (206) and from time-critical groups never seen in training (187) · same-pipeline in-set controls (258)